MATH1073: Calculus I Quiz# 2 Solution

1. The domain of definition for the function is $(-\infty, 0) \cup (0, \infty)$.

(i) find the intervals of increase or decrease;

$$f'(x) = -\frac{1}{x^2} + \frac{2}{x^3} = -\frac{x-2}{x^3}$$

Since $f'(x) = 0$ when $x = 2$ and does not exist when $x = 0$, the critical numbers are $x = 0$ and $x = 2$. The critical numbers devide the number line into three parts:

$$\begin{aligned} x < 0 &: f'(x) < 0, \text{decreasing} \\ 0 < x < 2 &: f'(x) > 0, \text{increasing} \\ x > 2 &: f'(x) < 0, \text{decreasing} \end{aligned}$$

(ii) find the local maximum and minimum values;
According to the answer in (a), $f(2) = 1.25$ is a local maximum and there is no local mininum.

(iii) find the intervals of concavity and the inflection points.
The second derivative:

$$f''(x) = \frac{2}{x^3} - \frac{6}{x^4} = \frac{2(x-3)}{x^4}$$

When $x < 0$ or $0 < x < 3$, $f''(x) < 0$, $f(x)$ is concave. When $x > 3$, $f''(x) > 0$, $f(x)$ is convex. $x = 3$ is the only inflection point of $f(x)$.

2. (i) For $y = x^3 + 3\ln x, \quad x > 0$.
First, we find the derivative of the function:

$$y' = 3x^2 + \frac{3}{x}$$

For $x > 0$, we have

$$3x^2 + \frac{3}{x} > 0$$

So the function is increasing on $(0, \infty)$.

(ii) For $y = 2\sin x + \sin 2x, \quad 0 \le x \le 2\pi$.
The derivative is:

$$y' = 2\cos x + 2\cos 2x$$

Setting the derivative equal to zero gives us:

$$2\cos x + 2\cos 2x = 0.$$

Note that $\cos 2x = 2\cos^2 x - 1$. So

$$2\cos x + 2(2\cos^2 x - 1) = 0$$

which gives $\cos x = -1$ or $\cos x = 1/2$. Therefore, $y' = 0$ when $x = \frac{\pi}{3}, \pi, \frac{5\pi}{3}$.
Moreover, on the intervals $(0, \frac{\pi}{3})$ and $(\frac{5\pi}{3}, 2\pi)$, $y' > 0$; on the intervals $(\frac{\pi}{3}, \pi)$ and $(\pi, \frac{5\pi}{3})$, $y' < 0$. We find that the function is increasing on $[0, \frac{\pi}{3}]$ and $[\frac{5\pi}{3}, 2\pi]$, and decreasing on $[\frac{\pi}{3}, \frac{5\pi}{3}]$.

3. (i) $f'(x) = 4x^3 + 54x^2 + 240x + 1, \quad f''(x) = 12x^2 + 108x + 240 = 12\,(x^2 + 9x + 20) = 12(x+4)(x+5)$. Thus, the important points are $x = -4$ and $x = -5$. For $x > -4, x+4$ and $x+5$ are positive, and, therefore, so is $f''(x)$. For $-5 < x < -4, \quad x+4$ is negative and $x + 5$ is positive; hence, $f''(x)$ is negative. For $x < -5$, both $x + 4$ and $x + 5$ are negative, and, therefore, $f''(x)$ is positive. Therefore, the function is convex for $x > -4$ and for $x < -5$. The function is concave for $-5 < x < -4$. Thus, the inflection points are $x = -4$ and $x = -5$.

(ii) $f(x) = \left[\left(x - \frac{1}{2}\right) + \frac{1}{2}\right] / \left[2\left(x - \frac{1}{2}\right)\right] = \frac{1}{2}\{1+[1/(2x-1)]\}$. Hence, $f'(x) = \frac{1}{2}\,[-1/(2x-1)^2]\cdot 2 = -1/(2x-1)^2$. Then $f''(x) = [1/(2x-1)^3]\cdot 2 = 2/(2x-1)^3$. For $x > \frac{1}{2}, \quad 2x-1 > 0, f''(x) > 0$, and the function is convex. For $x < \frac{1}{2}, 2x - 1 < 0, f''(x) < 0$, and the function is concave. There is no inflection point, since $f(x)$ is not defined when $x = \frac{1}{2}$.

4. (i) $\lim_{x\to 0} \frac{3^x - 2^x}{x} = \lim_{x\to 0} \frac{\ln 3\cdot 3^x - \ln 2\cdot 2^x}{1} = \ln 3 - \ln 2$.

(ii) $\lim_{x\to 1} \frac{\ln x}{\tan \pi x} = \lim_{x\to 1} \frac{1/x}{\pi \sec^2 \pi x} = \frac{1}{\pi(-1)^2} = \frac{1}{\pi}$

(iii) Note that $x \ln x = \frac{\ln x}{1/x}$, to which L'Hôpital's rule applies. Therefore,

$$\lim_{x\to 0^+} \frac{\ln x}{1/x} = \lim_{x\to 0^+} \frac{1/x}{-1/x^2} = -\lim_{x\to 0^+} x = 0.$$

(iv) Note that $y = (1-2x)^{1/x} \Rightarrow \ln y = \frac{1}{x}\ln(1-2x)$, so

$$\lim_{x\to 0} \ln y = \lim_{x\to 0} \frac{\ln(1-2x)}{x} \overset{\text{H}}{=} \lim_{x\to 0} \frac{-2/(1-2x)}{1} = -2 \Rightarrow$$
$$\lim_{x\to 0} (1-2x)^{1/x} = \lim_{x\to 0} e^{\ln y} = e^{-2}.$$

(v) $\lim_{x\to(\pi/2)^-} \frac{\sec x}{\tan x} \overset{\text{H}}{=} \lim_{x\to(\pi/2)^-} \frac{\sec x \tan x}{\sec^2 x} = \lim_{x\to(\pi/2)^-} \frac{\tan x}{\sec x}$. Repeated applications of l'Hospital's Rule result in the original limit or the limit of the reciprocal of the function. Another method is to simplify first:

$$\lim_{x\to(\pi/2)^-} \frac{\sec x}{\tan x} = \lim_{x\to(\pi/2)^-} \frac{1/\cos x}{\sin x/\cos x} = \lim_{x\to(\pi/2)^-} \frac{1}{\sin x} = \frac{1}{1} = 1$$

5. Since $\lim_{h\to 0}[f(x+h) - 2f(x) + f(x-h)] = f(x) - 2f(x) + f(x) = 0 \quad [f$ is differentiable and hence continuous $]$ and $\lim_{h\to 0} h^2 = 0$, we can apply l'Hospital's Rule:

$$\lim_{h\to 0} \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} \overset{\text{H}}{=} \lim_{h\to 0} \frac{f'(x+h) - f'(x-h)}{2h}.$$

Since $\lim_{h\to 0}[f'(x+h) - f'(x-h)] = f'(x) - f'(x) = 0 \quad [f'$ is differentiable and hence continuous $]$ and $\lim_{h\to 0} h = 0$, we can apply l'Hospital's Rule again:

$$\lim_{h\to 0} \frac{f'(x+h) - f'(x-h)}{2h} \overset{\text{H}}{=} \frac{f''(x) - (-f''(x))}{2} = f''(x).$$